Design Milestones of Coreguard

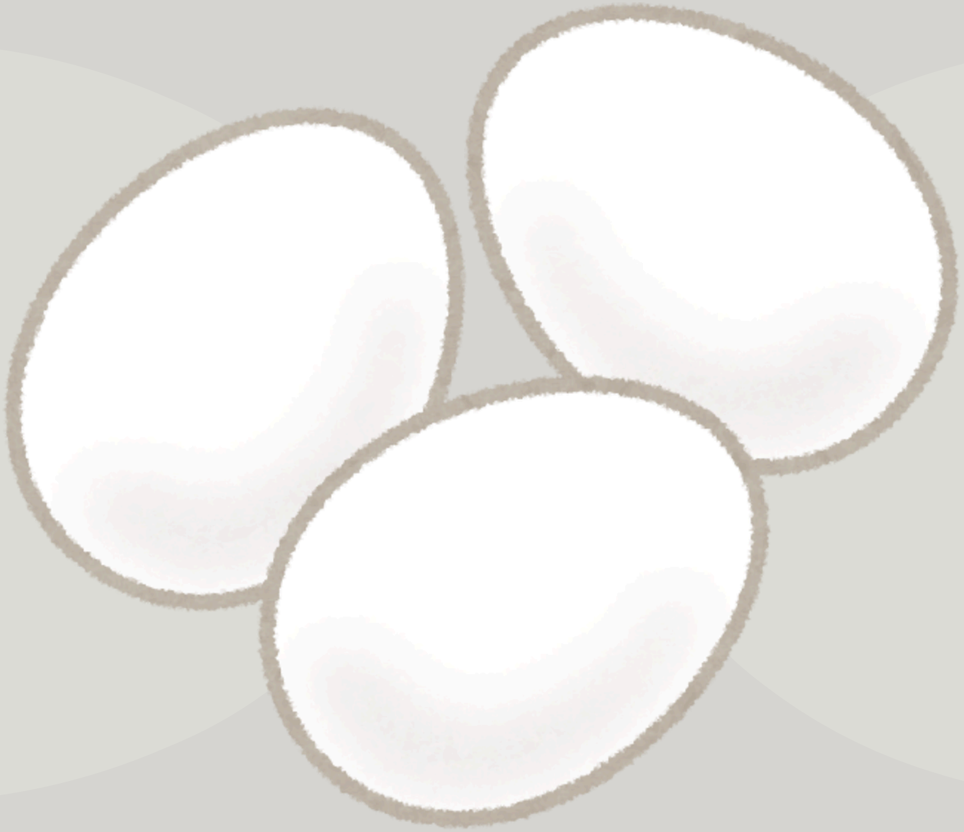
IDE 102.A : DESİGN AS CREATİVE PROBLEM SOLVING (SPRİNG 2025/26)

BESTE YILMAZ

Project Overview

the goal:

To develop a more balanced system with using ai as a tool that can protect the egg from a 3-meter drop and absorb the impact better with limited products.



the limitations:

- max 20 straws
- 1 meter tape
- 2 meters string
- structure must fit inside 15 cm cube

Egg Drop 1 analysis

Description

The first prototype used:

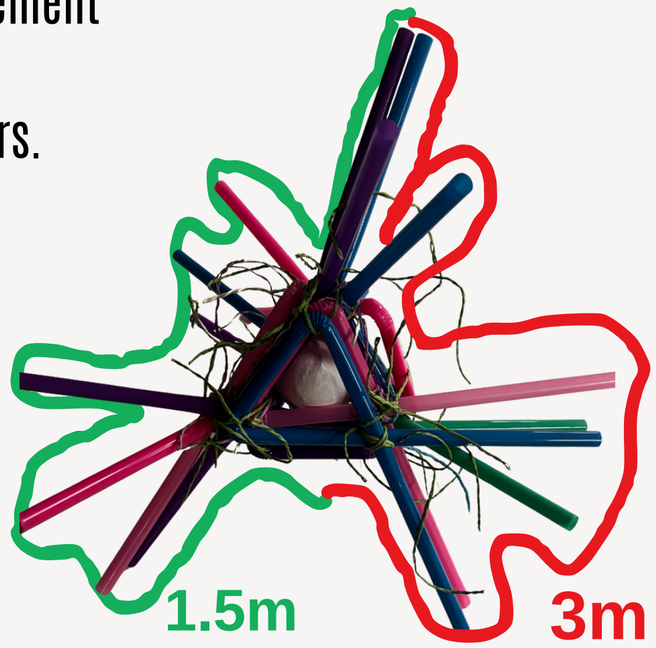
- radial outer straws
- triangular inner structure
- central egg placement

Observation

Egg survived 1.5 meters.

Problems

- asymmetry
- possible rotation
- weak knots



Egg Drop 2 (coreguard) analysis

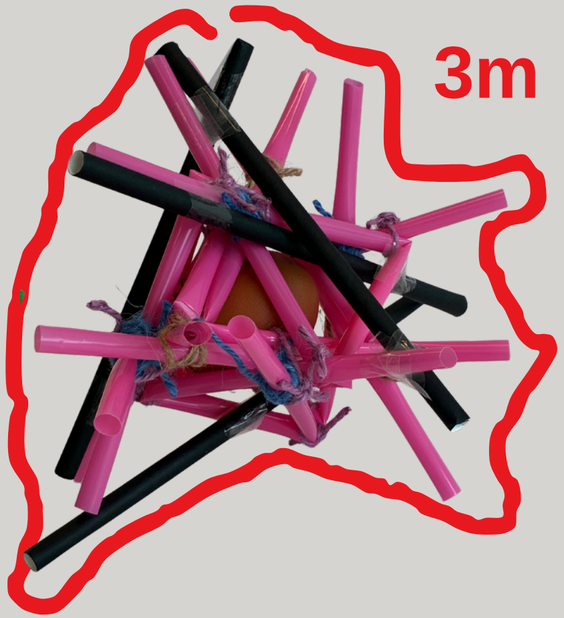
Instead of rebuilding the structure, I decided to improve my existing system.

Design choices:

- keep triangular core
- keep radial outer arms
- allow controlled bending
- maintain lightweight structure

Reason:

Impact absorption is more important than perfect symmetry.



Improving Coreguard

AI Part of the Process

AI Role Tag

Clarifier

Prompt

"If my structure fails at 3 meters, where will it most likely fail first?"

AI Insight

The first failure would likely occur in the outer straws because they receive the first impact.
Another weak point could be the knots, since strong impact can move the connections.
This helped me understand where the structure needed improvement.

AI Role Tag

Expander

Prompt

"I want to see geometric structures that are more stable than my current design."

AI suggested several structural ideas:

- Octahedron structure
- Tensegrity structure
- Double ring structure
- Symmetrical cube structure

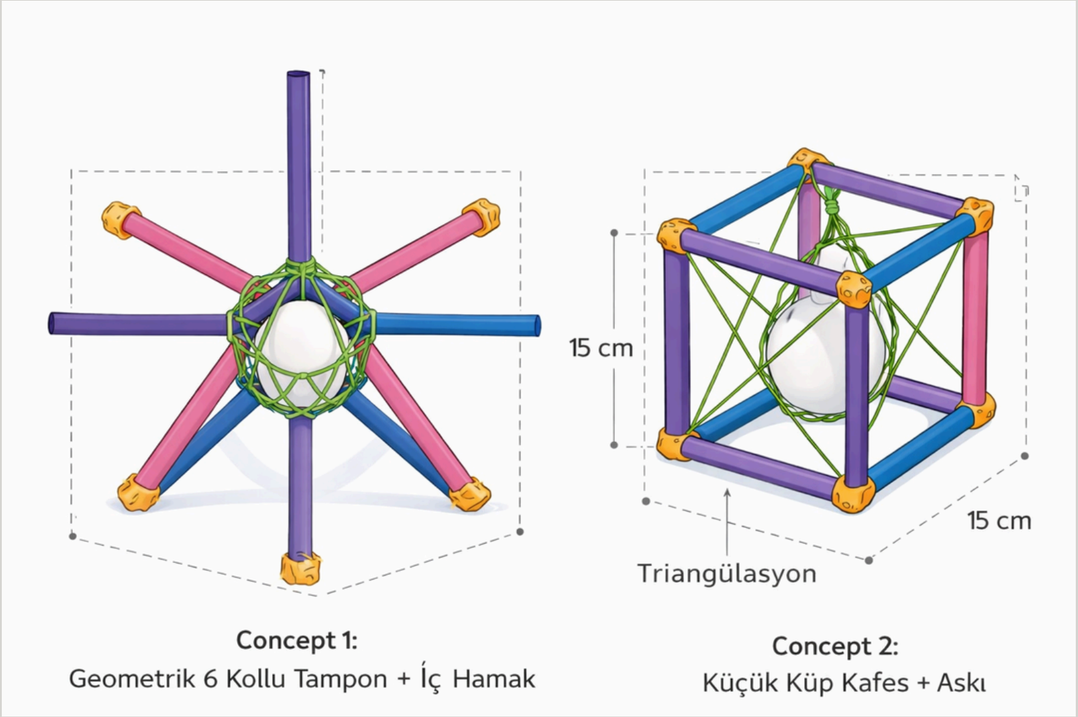
These concepts focused on symmetry and distributing the impact force.

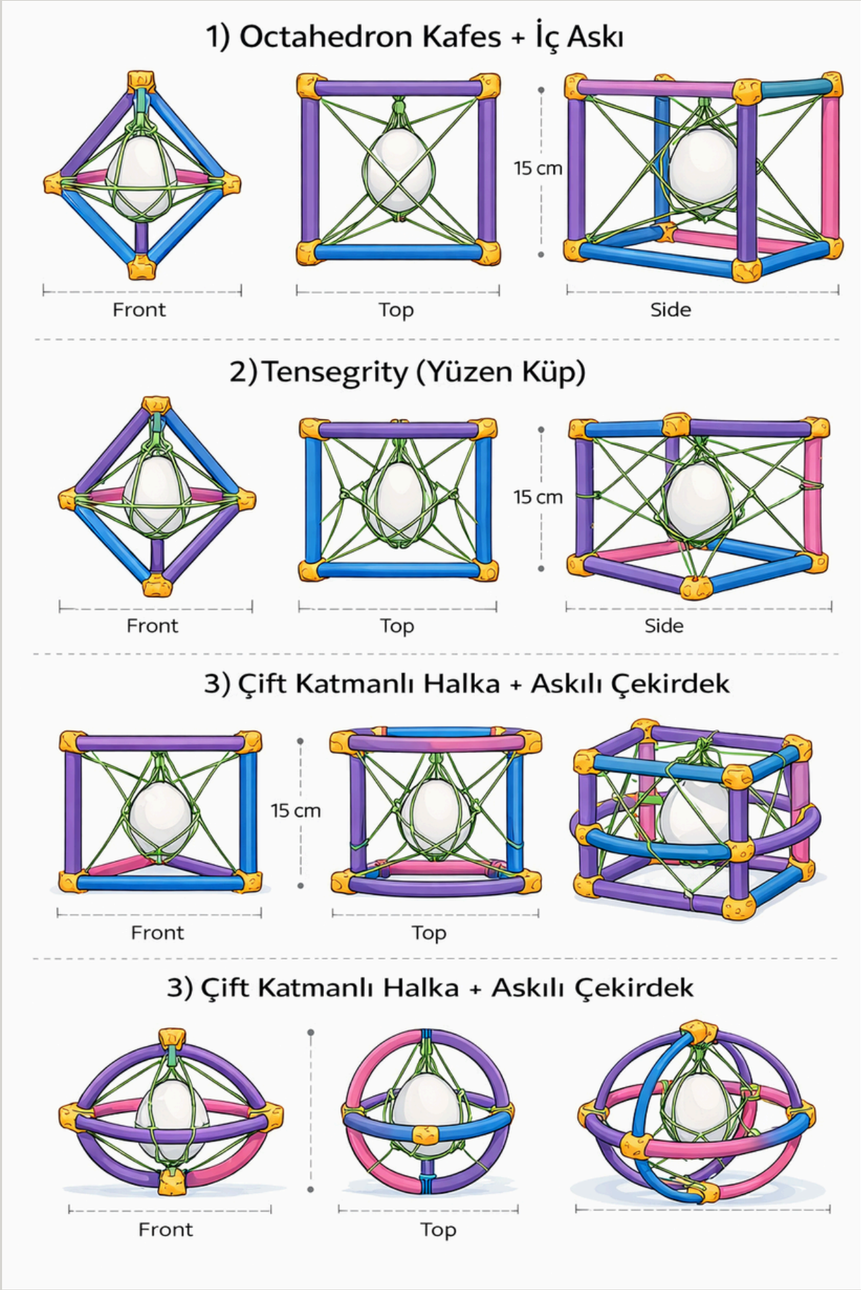
AI Visual Output:

These visual concepts were generated during the AI consultation phase to explore alternative structural systems.The goal was to understand how different geometries could improve stability and distribute impact forces during the drop.

Although these structures were not directly used in the final prototype, they helped evaluate possible design directions and compare them with the existing triangular-based system.

This process supported the decision to improve the current structure instead of building a completely new one.





Rejected Ideas of AI

During the AI process, several structural ideas were suggested. However, some of them were not practical for my materials and construction method.

Tensegrity Structure

This system depends on string tension. However, it is very difficult to keep all strings at the same tension when building by hand. Even a small knot movement could change the balance of the structure. For this reason, I did not consider this system reliable.

Double Ring Structure

This design tries to spread the impact force using circular rings. However, straws are not easy to bend into smooth circles. This system would require many connections, and more connections mean more weak points.

Perfect Octahedron Geometry

This geometry is very stable in theory, but it requires straws of exactly the same length. In hand-made construction, small differences can happen, and these small errors could affect the balance of the structure

Accepted and Transformed Ideas

After working with AI, I accepted some ideas and adapted them to my own design.

Accepted Ideas

Triangular Stability

I learned that triangle shapes are more stable than square structures. Because of this, I kept the triangular inner core in my design.

Radial Impact Absorption

The outer straws take the first impact when the structure hits the ground. This helps protect the egg from direct contact with the ground.

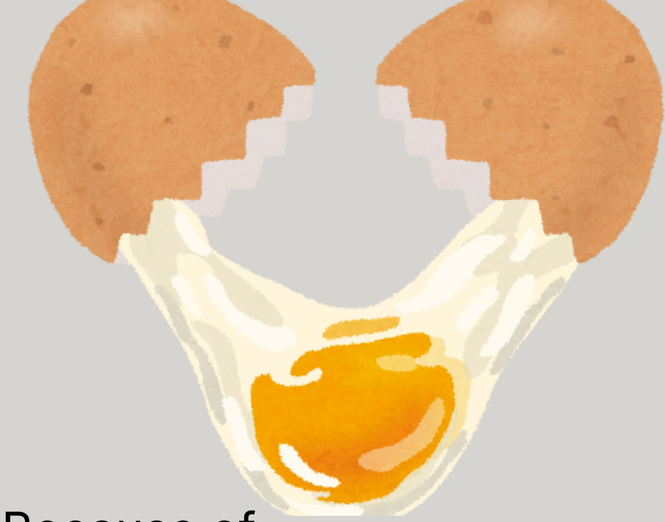
Transformed Ideas

AI suggested very symmetrical structures. Instead of copying them exactly, I adapted the ideas to my own design.

- I kept the outer straws to absorb impact
- I allowed the structure to bend slightly
- I avoided making the structure too rigid

This approach helped slow down the impact and spread the energy.

Why did it Failed?



Structural Asymmetry

The structure was not perfectly symmetrical. Because of this, the structure may have rotated during the fall and the impact may have concentrated on one side.

Weak Connections

The nodes were connected with string and tape. During a strong impact, some of these connections may have moved and changed the shape of the structure.

Limited Energy Absorption

The outer straws absorbed some of the impact, but it may not have been enough for a 3 meter drop. The force could still reach the inner structure and the egg.

Reflection

After the first test from 1.5 meters was successful, I chose to continue developing the **same triangular-based structure** instead of starting a completely new design. I believed the triangular core already provided **basic stability**, so improving this system seemed like a more realistic approach.

Working with AI helped me explore different structural ideas and compare them with my own design. Because of this, I was able to eliminate some options more confidently and focus on the ideas that could work with my materials and constraints.

During this process, I also realized how important **symmetry** is. Even small differences in length or balance can change how the structure falls and how the impact force spreads. This made me more aware of how careful I need to be when building and adjusting the structure.